

2. Legionnaires’ disease is a serious lung infection caused by *Legionella* bacteria.

- (a) In 2015 there were 18 confirmed cases of Legionnaires’ disease in Wales. The population of Wales in 2015 was 3 099 100.

Calculate the percentage of the population of Wales that suffered from Legionnaires’ disease in 2015. Express your answer in **standard form**. [2]

Answer =

- (b) *Legionella* is a Gram negative, bacillus bacterium that can live in domestic water supplies. If small droplets of contaminated water are inhaled the bacteria can cause pneumonia-like symptoms.

State what is meant by a Gram negative, bacillus bacterium. [2]

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- (c) During an outbreak, a sample of contaminated water was studied to find which antibiotic was most effective. The three antibiotics were erythromycin, ciprofloxacin and azithromycin. A range of concentrations of each antibiotic were used.

A microbiologist carried out the investigation as follows:

- Sterile nutrient agar plates were set up containing different concentrations of erythromycin.
- 0.5 cm³ of diluted contaminated water was transferred onto each plate.
- The same process was repeated for ciprofloxacin and azithromycin.

The results following incubation are shown in the table.

Concentration of antibiotic / %	Number of colonies on plate		
	erythromycin	ciprofloxacin	azithromycin
0	86	86	86
10	70	43	0
20	47	22	0
30	32	0	0
40	24	0	0



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- (i) The microbiologist had diluted the original sample of contaminated water by a factor of 10^{-3} . Use the information given to calculate the number of *Legionella* bacteria in 1 cm^3 of the original contaminated sample. Show your working. [2]

Number of *Legionella* bacteria in 1 cm^3 =

- (ii) Suggest a reason for the following:
- I. When the microbiologist diluted the original sample by a factor of 10^{-2} she could not calculate the number of *Legionella* bacteria per cm^3 . [1]

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- II. When she diluted the original sample by 10^{-6} a lower number of *Legionella* bacteria per cm^3 was calculated. [1]

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- (d) The cost of ciprofloxacin and azithromycin are approximately the same per unit mass. Use the results of this experiment to suggest and explain which antibiotic would most likely to be recommended by the Welsh Government to treat this disease. [2]

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